CENTRE CODE				CANDIDATE NUMBER

Name _____

MUCHINGA PROVINCE MOCK EXAMINATION

DESIGN AND TECHNOLOGY DEPARTMENT

GRADE 12 Paper 1 (theory and design) 6045/2

10th July, 2024 Duration 2h 30min Total Marks: 110

Instructions to candidates

Write your name, centre number and candidate number in the spaces at the top of this page and on the separate answer booklet provided.

There **two** sections in this paper.

Section A

Instruction to the candidates

1. Write your name, centre number and your examination number on every page in this question paper and on the separate answer booklet provided.
2. If working is needed for any question, it must be shown in the space provided in the question paper answer booklet.
3. There are two0 sections in this paper. Section A and B. Answer all questions in section A. Write your answers in the spaces provided in the question paper.
4. Section b has three parts I, II and III. Part I is compulsory.
5. Answer one question in part II and one question in part III.

Write your answers in answer booklet provided.

6. The question in part III must be answered on A3paper drawing paper provided
7. Where free hand sketch is required, it must be drawn to half-full size with good line work
8. at the end of the examination, fasten the answer booklet and drawing paper used securely to the question paper.

Information for Candidates

The number of marks is given in brackets [ ] at the end of each question.

All dimensions are in millimetres

Cell phones and calculator are not allowed in the examination room

For candidate use	**For examiner's use**
Section A	
Section B	
TATOL	

SECTION A [30 MARKS]

Answer all questions from this section in the spaces provided in the question paper.

1. (a) Fig. 1 shows four items used to fasten materials together.

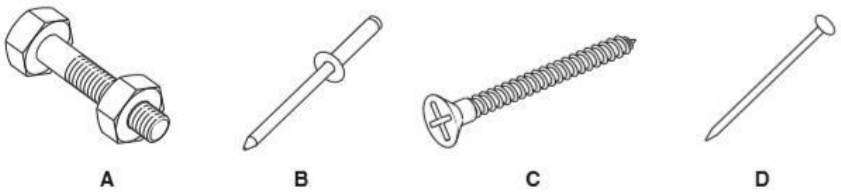


Fig 1

Name each of the items **A, B, C** and **D** shown in Fig. 1.

A _____

B _____

C _____

D _____ **[4]**

(b) Fig. 2 shows two safety clothing [PPE] signs



Fig 2

Describe a workshop process where:

i **i.** a mask must be worn; _____

ii ii. gloves must be worn. _____ **[2]**

2. (a) **Fig. 3** shows three examples of a lever.

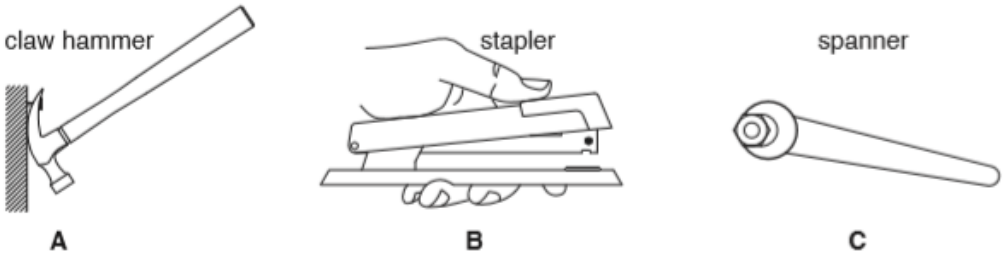


Fig 3

i. State the order of lever used in each example.

Lever A _____

Lever B _____

Lever C _____ **[3]**

ii. State which of the three levers will not provide a mechanical advantage to the user _____ **[1]**

(b) Chanda's company specializes in advertising different products on behalf of the business communities. Suggest two methods of advertising that she can use.

_____

_____ **[2]**

3. **(a) Fig. 4** shows a saucepan.



Fig 4

i. Name a thermosetting plastic that could be used for the handle.

_____ **[1]**

ii. Give a reason for your choice. _____

_____ **[1]**

iii. Name a ferrous alloy that could be used for the saucepan body.

_____ **[1]**

iv. Give a reason for your choice.

_____ **[1]**

(b) Give one example of each of the energy sources named below.

i. Renewable energy_____ **[1]**

ii. Fossil fuel _____ **[1]**

4. Figure 5 shows a circuit diagram for a system whose solenoid is controlled by a light sensitive circuit.

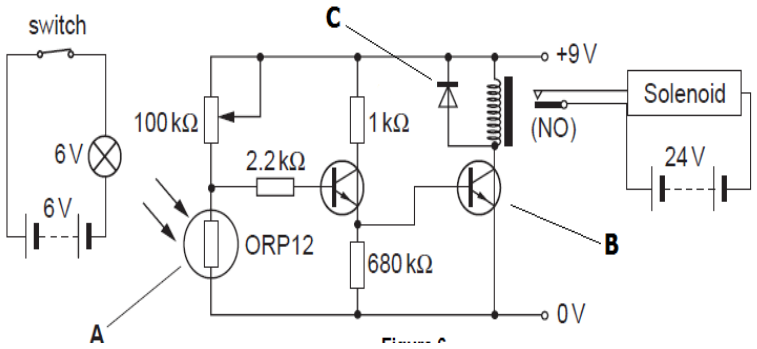


Fig 5

i. Name components **A, B** and **C.**

A _____

B _____

C _____ **[3]**

ii. Explain the purpose of component C in the circuit._____

_____ **[2]**

iii. What is the meaning of a symbol (NO) shown on the solenoid?

_____ **[1]**

5. a) Figure 6 below shows a square metal frame with horizontal force acting on it.

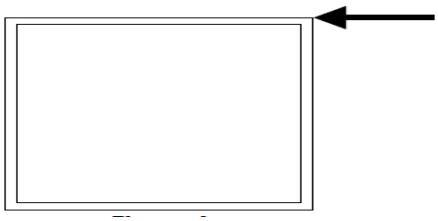


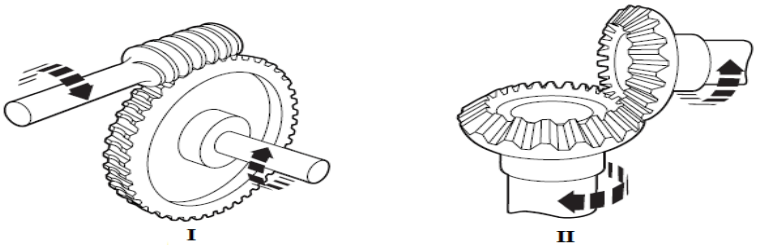
Fig 6

Name two methods one would use to make the frame rigid.

Method 1: _____

Method 2: _____ **[2]**

Figure 7 shows two types of gear system that transfer rotary motion through 90^0.



Name the two types of gear system shown.

i _____

ii. _____ **[2]**

Give one advantage of type I over type II

_____ **[2]**

Section B [80 marks]

There are three parts in this section Part I, II and III. Part I is compulsory. Answer one question from Part II and one question from Part III.

Part I Compulsory.

6. Figure 8 shows a situation obtaining in the HOD's office at your school. Study and refer to it to answer the questions that follow.

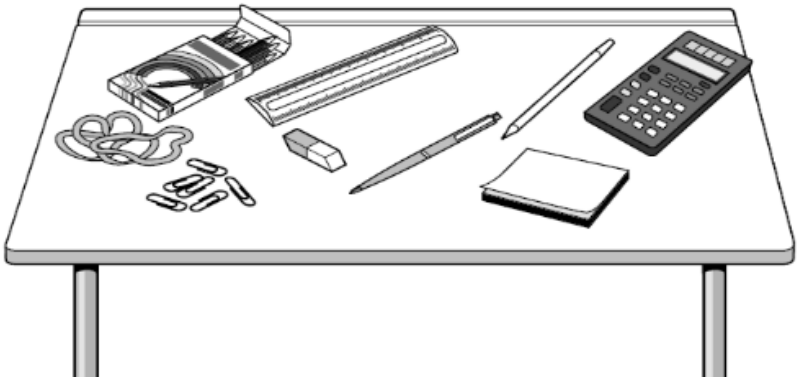


Fig 8

i Write down the situation. [3]

ii Identify the problem from the situation you have stated above. [2]

iii Derive a design brief from the problem in a) (ii). [3]

iv Using annotations generate one well rendered possible solution in pictorial view for the situation above. [12]

v Identify one advantage and one disadvantage from the possible solution. [2]

Mention three research tools that you can use to gather more information. [3]

[Total: 25 marks]

Part II has three questions. Answer one question only.

7. Fig 9 shows two different types of structure used to support wind turbines.

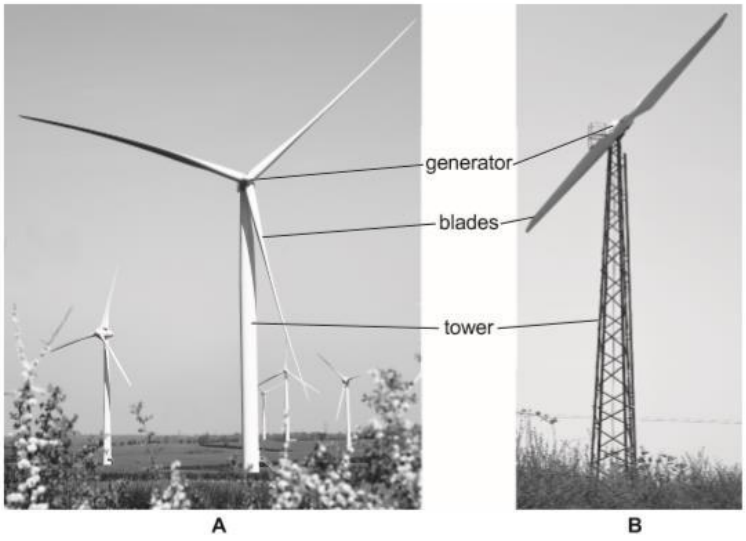


Fig 9

(i) State the type of structure used for the tower on each of the turbines. [2]

(ii) Name one stationary load that will act on the base of each tower when the blades are at rest. [1]

(iii) Name two moving loads that will act on the wind turbines when they are generating. [2]

(iv) Explain how the turbine towers in Fig. 10 maintain equilibrium. [2]

(b) Use sketches and notes to describe the differences between the following structural members.

(i) Beam [2]

(ii) Strut [2]

(iii) Tie [2]

(c) State three forces that may affect the stability of the structure when a person sits on the chair. [3]

(d) Give two reasons why a designer would need to reduce friction in a mechanism. [2]

(e) State two methods that can be used to reduce friction in a mechanism. [2]

(f) Sketch a mortice gauge and explain its purpose. [5]

[Total: 25marks]

8. a). Figure 10 shows the basic design of a child's toy. As it is pulled and pushed along, the driver moves up and down.

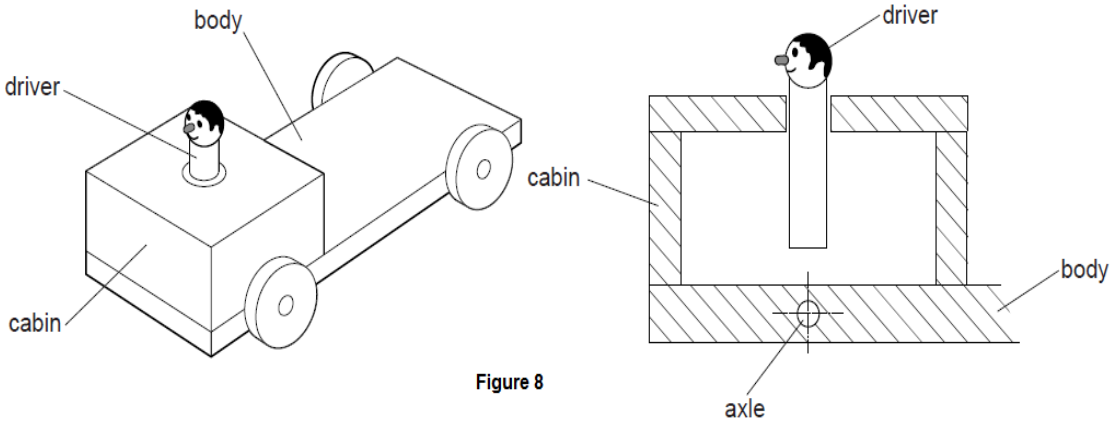


Figure 8

I. Draw a mechanism that will make the driver move up and down when pulled along. [3]

II. Name the mechanism you have drawn in i above. [1]

III. Draw a mechanism that will make the driver rise slowly and fall rapidly when pulled along. [3]

IV. Name the mechanism you have drawn in iii above. [1]

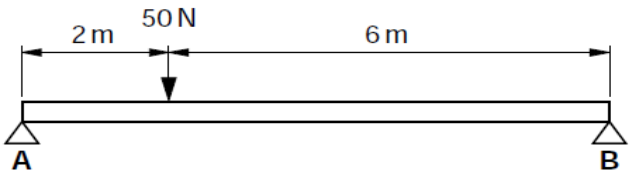
V. Define the term mechanism. [2]

b). Use drawings to show each of the following types of motion.

I. Oscillating motion [2]
II. Linear motion [2]
III. Reciprocating motion [2]

[Total: 25 marks]

c) Figure 11 shows a simple beam supported at A and B with a load of 50N at the point indicated.



Calculate the values of the reactions at A and B. [3]

Define the following terms:

i. Fulcrum/pivot

ii. Effort

iii. Load [6]

[Total: 25 marks]

9) a) Figure 12 shows views of a child's table.

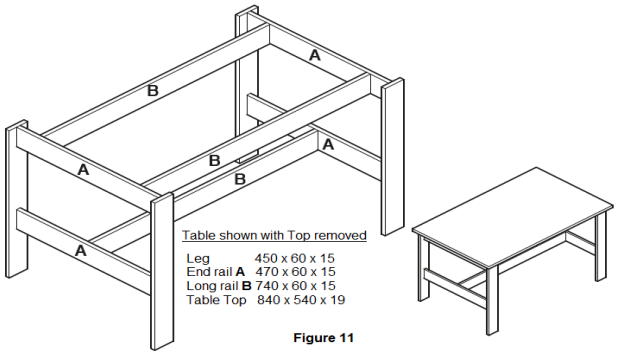


Figure 11

i. Using the information above, draw and complete a cutting/part list with the following headings:

Part
Quantity
Length
Width
Thickness [20]

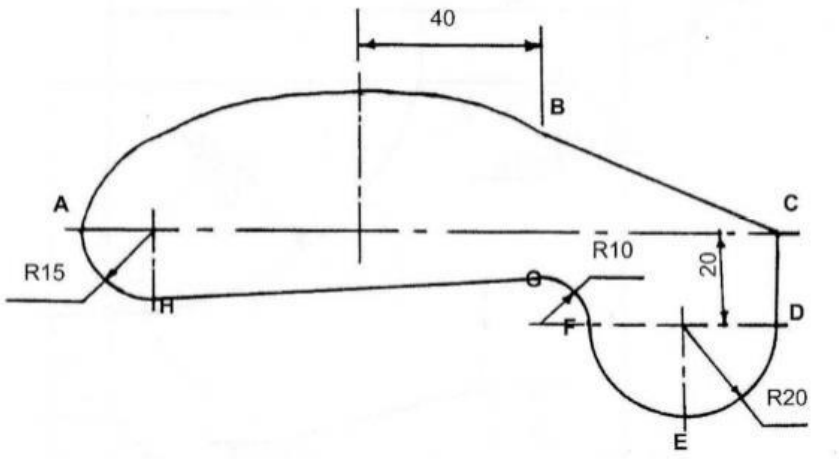
ii. Suggest one type of a manufactured board that can be used for a table top. [1]

iii. Briefly explain four advantages manufactured boards have over solid wood. [4]

[Total: 25 marks]

Part III has two questions. Answer one question on the A3 paper provided.

10. Fig 13 shows the profile of a garden. AB is elliptical with major axis 120mm and minor axis 60mm. straight line BC is tangential at B. using geometrical methods, construct the profile of the garden in full size



[Total: 30 marks]

a) Mpika day secondary school has a canteen that serves meal to members of staff during lunch time. You have been asked to design models that would contain a salt seller (cruets) and a bottle of chilli to be used on each table in the canteen. The containers could be of any shape as long as they are not cylinders.

i Sketch a model for proposed design in pictorial view. [3]

ii. Construct the surface development of your prosed design model. [12]

iii. Suggest the name of the canteen and print it on your model. [3]

iv. Differentiate using sketches by indicating the container for salt and chill. [12]

[Total: 30 marks]

11. **Fig 14** shows a design for a part of a chicken feeder with a pentagonal pyramid base as shown below.

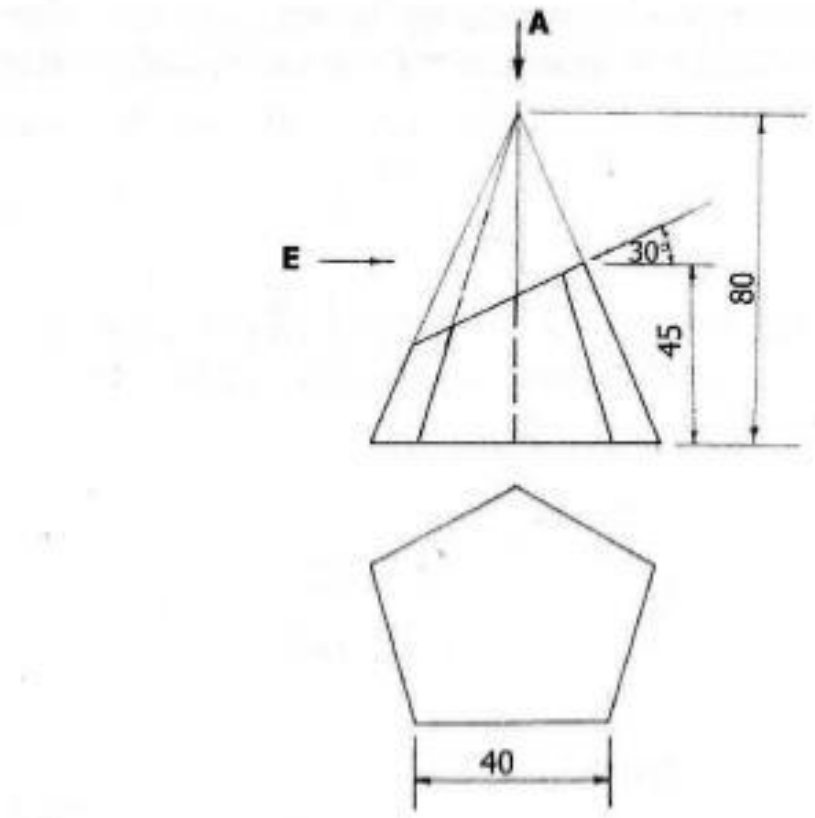


Fig 14

i Draw the complete plan.
ii Draw the front elevation.
iii Draw the surface development of the rectangular base only seam along 30° cut.

[Total: 30 marks]